Are Machine Learning Improvements Statistically Real? Dependence-Aware Inference for Reliable Model Comparison on Spatiotemporal Data

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1 Introduction

Empirical machine learning advances one comparison at a time. A new gradient-boosting configuration is reported to beat a random forest, a deep architecture to beat gradient boosting, a feature set to beat another. Such claims rest on a small number of summary statistics (an average accuracy, an F1-score, a root mean squared error) computed by cross-validation, and, in the better cases, on a statistical test or a confidence interval for the difference. The test is meant to answer a simple question: *is the observed improvement larger than what the randomness of the data and of the validation procedure could produce on its own?*

The standard tools for that question [1–4] were developed for observations that are independent and identically distributed (i.i.d.). Much of the data on which machine learning is now used in practice is not. Meteorological, environmental, agricultural, epidemiological and mobility data are collected at locations and times that are close to one another, and nearby observations share unmeasured drivers. Two consequences are well documented. First, random K-fold cross-validation places near-duplicates of test observations in the training set, so that it measures interpolation rather than prediction at new places or times, and is optimistic when the latter is the target [5–8]. Second, and much less studied, the *errors* of a fitted model are themselves dependent, so that n test observations carry much less information than n independent ones.

The first problem has a well-known remedy: blocked (spatial, temporal or spatio-temporal) cross-validation [5, 9, 10]. The second problem is the subject of this paper. Blocking changes *what* is estimated; it does not tell us *how precisely* it is estimated. A practitioner who computes a blocked cross-validation score for two models and then applies a paired t-test, a bootstrap that resamples observations, or a test on fold-wise scores still relies on independence assumptions that the data violate. Whether an apparent improvement is "statistically real" then depends as much on these hidden assumptions as on the models.

Our starting point is the per-observation out-of-fold *loss differential* $d_i = \ell(y_i, \hat{f}_A(x_i)) - \ell(y_i, \hat{f}_B(x_i))$ between two models A and B. The mean of the d_i estimates the expected gain of A over B, and the comparison is a problem of inference on the mean of a dependent random field. Two sources of dependence must be handled at once. (i) Test-point dependence: the d_i inherit the spatial and temporal autocorrelation of the residuals of both models. (ii) Training dependence: all observations of a fold are predicted by the same fitted models, and the models of different folds are fitted on overlapping training sets. Existing procedures handle one source at most. Heteroskedasticity- and autocorrelation-consistent (HAC) variance estimators, as in the Diebold–Mariano test for forecasts [11] and its spatial extension [12], handle (i) for fixed predictors. The corrected resampled t-test of Nadeau and Bengio [2] handles (ii) for i.i.d. data.

Contributions.

The paper makes four contributions.

1. We formalise model comparison under spatio-temporal dependence as inference on the mean of a loss-differential field with a fold-level random effect, and we show (Proposition 1) that the widely used paired t-test on fold scores underestimates the variance by exactly the average covariance between fold means. This explains why it fails under random cross-validation even though it "looks" conservative with only $K - 1$ degrees of freedom.
2. We propose *DA-HAC*, a dependence-aware HAC procedure for comparing two models evaluated by blocked cross-validation. It combines (a) a product-Bartlett kernel estimator of the test-sampling variance whose spatial and temporal bandwidths are *calibrated from the empirical variogram of the loss differentials themselves*; (b) a method-of-moments estimate of the training-induced fold variance, inflated by a Nadeau–Bengio factor; and (c) a Welch–Satterthwaite reference distribution whose degrees of freedom reflect the effective sample size. All three steps are computed from a single cross-validation run.
3. We assess the validity of conventional and dependence-aware procedures (paired t, i.i.d. bootstrap, corrected resampled t, block bootstrap, linear mixed model, Diebold–Mariano–HAC and DA-HAC) in two simulation studies with exact control of the truth: one in which loss-differential fields are generated directly, and one with gradient-boosting learners trained on spatio-temporal data in which two models have equal expected loss by an exchangeability argument.
4. We apply all procedures to three public data sets (hourly weather in New York, hourly pedestrian counts in Melbourne, and yield-monitor maize data in Argentina),

including a "seed-null" diagnostic in which a model is compared with itself, and we derive practical recommendations. An open-source Python implementation is provided.

The emphasis of the paper is thus not on whether spatial or temporal validation is preferable, a question already addressed in the literature, but on the *validity of the inference* drawn from a given validation design. In the language of the call for this special issue, it is a contribution to measuring the trustworthiness of comparisons between machine learning models in real applications.

The rest of the paper is organised as follows. Section 2 reviews related work. Section 3 sets up the problem and analyses conventional tests. Section 4 presents DAHAC. Sections 5 and 6 report the simulation studies, Section 7 the case studies, and Section **??** discusses recommendations and limitations.

2 Related work

Statistical comparison of learning algorithms.

Dietterich [1] showed that the paired t-test on resampled hold-out scores has an inflated type I error rate, because training sets overlap, and proposed the 5×2 cross-validation test. Nadeau and Bengio [2] derived the corrected resampled t-test, which inflates the variance of repeated train/test differences by the factor $1/J + n_{\text{test}}/n_{\text{train}}$. Bengio and Grandvalet [13] proved that no unbiased estimator of the variance of K-fold cross-validation exists, which is why every practical procedure relies on a working model or a heuristic. Bouckaert and Frank [14] studied the replicability of these tests, Demšar [3] and Benavoli et al. [4] treated comparisons over multiple data sets, frequentist and Bayesian. More recently, Bayle et al. [15] established central limit theorems for K-fold cross-validation under algorithmic stability, and Bates et al. [16] clarified that cross-validation estimates the average error of the learning *procedure* rather than that of the fitted model, and proposed nested cross-validation intervals. Bouthillier et al. [17] quantified the sources of variance (data split, initialisation, hyperparameter search) in machine learning benchmarks, and Varoquaux [18] showed how wide cross-validation error bars are in small samples. All these works assume independent observations.

Validation under spatial and temporal dependence.

In ecology, remote sensing and the geosciences, a large literature shows that random cross-validation is optimistic for spatially structured data and recommends spatial blocking, buffering or nearest-neighbour distance matching [5–7, 9, 10, 19, 20]. Wadoux et al. [21] argued that, when the goal is the accuracy of a map over a fixed region, design-based probability sampling is preferable to any form of spatial cross-validation. In time series, Bergmeir and Benítez [8] compared blocked and out-of-sample schemes and Bergmeir et al. [22] showed when standard cross-validation remains valid for autoregressive models. These works are concerned with the *point estimate* of predictive performance and its bias; the uncertainty of the comparison between two models, which is the object of this paper, is rarely addressed.

Comparing predictions of dependent data.

The Diebold–Mariano (DM) test [11], with the small-sample correction of Harvey et al. [23], compares the expected loss of two forecasts by applying a HAC (Newey–West [24]) variance estimator to the loss-differential series. Hering and Genton [12] extended it to spatial fields, estimating the covariance of the loss differential with a parametric variogram. HAC estimation under cross-sectional dependence goes back to Conley [25] and, for panels, to Driscoll and Kraay [26]; bandwidth choice and the poor size of HAC tests in small effective samples motivated the fixed-b theory of Kiefer and Vogelsang [27]. Block and stationary bootstraps [28–30] provide resampling alternatives. Two features distinguish our setting from that of the DM test: the predictions are produced by learners that are *refitted* on each training fold, which adds a training-induced variance component absent from forecast comparisons; and the dependence ranges of the loss differential are unknown and must be estimated without a parametric model of the data. DA-HAC addresses both.

Trustworthy machine learning and this special issue.

Recent contributions in this journal have advocated statistical tools for the robustness of machine learning models [31], practical safeguards against overfitting [32], and effect-size reporting beyond p-values in large data sets [33]. The present paper complements them by addressing the validity of the comparison step itself when observations are dependent.

3 Problem formulation

3.1 Setting and estimand

We observe n labelled examples $z_i = (x_i, y_i)$, $i = 1, \dots, n$, each attached to a spatial location $s_i \in \mathbb{R}^p$ (typically $p = 2$) and a time $t_i \in \mathbb{R}$; either component may be absent. Two learning procedures A and B (for instance, XGBoost and a random forest, or one learner with two feature sets) are to be compared with a loss ℓ, such as the squared error or the logarithmic loss [34].

The observations are grouped into G spatio-temporal blocks (for instance, square tiles crossed with time windows) and the blocks are allocated to K folds $F_1, \dots, F_K$, with $n_k = |F_k|$. For each fold k, both learners are fitted on a training set $T_k \subseteq \{i \notin F_k\}$, from which observations within a buffer distance of the fold may be removed, and are used to predict the observations of F_k. With $k(i)$ the fold of observation i, the out-of-fold *loss differential* is

$$d_i = \ell\big(y_i, \hat{f}_A^{(k(i))}(x_i)\big) - \ell\big(y_i, \hat{f}_B^{(k(i))}(x_i)\big), \tag{1}$$

so that negative values favour A, and the cross-validated estimate of the gain is $\hat{\Delta} = \bar{d} = n^{-1} \sum_i d_i$.

Definition 1 (Estimand) The *expected cross-validated loss differential* is $\Delta_n = \mathbb{E}[\hat{\Delta}]$, where the expectation is over the data-generating process and the randomness of the learners. It

is the difference in expected loss between the two learning procedures, trained on about $n(1-1/K)$ observations, when predicting observations that are separated from the training data as prescribed by the blocking design.

The estimand depends on the validation design, as stressed by Bates et al. [16] for i.i.d. data and by the spatial validation literature [5, 21]: with random folds it describes interpolation, with spatial blocks it describes extrapolation to new areas. We take the design as given, chosen to match the intended use of the model, and focus on testing $H_0 : \Delta_n = 0$ against $H_1 : \Delta_n \neq 0$ and on confidence intervals for Δ_n.

3.2 A working model for loss differentials

The difficulty is that the d_i are dependent for two distinct reasons. We make them explicit with the following decomposition.

Definition 2 (Working model)

$$d_i = \Delta_n + \eta_{k(i)} + u_i, \tag{2}$$

where $(\eta_1, \dots, \eta_K)$ are exchangeable fold effects with $\mathrm{Var}(\eta_k) = \tau^2$ and $\mathrm{Corr}(\eta_k, \eta_l) = \rho$ for $k \neq l$, independent of u, and $u = (u_i)$ is a zero-mean, second-order stationary field with $\mathrm{Cov}(u_i, u_j) = C(s_i - s_j, t_i - t_j)$.

The fold effect η_k captures the variability induced by fitting the two learners on the training set T_k: an unlucky training sample, a different random seed or a different tree structure shifts the losses of all observations of the fold in the same direction. Because the training sets of different folds share a fraction $(K-2)/(K-1)$ of their observations, the fold effects are positively correlated. The field u captures test-sampling variability; it inherits the spatial and temporal autocorrelation of the residuals of both models, which is substantial whenever both models miss a spatially structured signal. Under (2),

$$\mathrm{Var}(\hat{\Delta}) = \underbrace{\frac{1}{n^2}\sum_{i=1}^{n}\sum_{j=1}^{n} C(s_i - s_j, t_i - t_j)}_{V_u} + \underbrace{\tau^2\Big[\sum_k \frac{n_k^2}{n^2} + \rho \sum_{k \neq l} \frac{n_k n_l}{n^2}\Big]}_{V_\eta}. \tag{3}$$

With equal fold sizes, $V_\eta = \tau^2[1/K + \rho(1-1/K)]$. Writing $V_u = \sigma_u^2 / n_{\mathrm{eff}}$ defines the *effective sample size* $n_{\mathrm{eff}} = n/[1 + (n-1)\bar{\rho}_u]$, where $\bar{\rho}_u$ is the average correlation between distinct u_i. Under moderate dependence n_{eff} can be orders of magnitude smaller than n.

3.3 Why conventional procedures fail

Procedures that assume independent observations.

The i.i.d. bootstrap of the d_i, the z-test on the d_i and McNemar-type tests estimate σ^2/n and therefore ignore both terms of (3) beyond the diagonal. Their variance is too small by the design-effect factor n/n_{eff}, and by V_η even without spatial dependence.

The paired t-test on fold scores.

The most common procedure computes the K fold means $m_k = n_k^{-1} \sum_{i \in F_k} d_i$ and refers $\bar{m}/(s_m/\sqrt{K})$ to a t_{K-1} distribution, with s_m^2 the sample variance of the m_k. Its weakness is captured by the following identity.

Proposition 1 *Let $m_1, \dots, m_K$ have common mean, average variance $\bar{v} = K^{-1} \sum_k \operatorname{Var}(m_k)$ and average pairwise covariance $\bar{c} = [K(K-1)]^{-1} \sum_{k \neq l} \operatorname{Cov}(m_k, m_l)$. Then*

$$\mathbb{E}\Big[\frac{s_m^2}{K}\Big] - \operatorname{Var}(\bar{m}) = -\bar{c}.$$

Proof $\operatorname{Var}(\bar{m}) = K^{-2}\big[\sum_k \operatorname{Var}(m_k) + \sum_{k \neq l} \operatorname{Cov}(m_k, m_l)\big] = [\bar{v} + (K-1)\bar{c}]/K$. Since the means are equal, $\mathbb{E} \sum_k (m_k - \bar{m})^2 = \sum_k \operatorname{Var}(m_k) - K \operatorname{Var}(\bar{m}) = (K-1)(\bar{v} - \bar{c})$, hence $\mathbb{E}[s_m^2]/K = (\bar{v} - \bar{c})/K$. Subtracting gives the result. □

The paired t-test is therefore valid only if fold means are uncorrelated. Under random K-fold cross-validation of dependent data, every fold is spread over the whole domain, the folds share the same spatial and temporal fluctuations, and $\bar{c}$ is close to $\operatorname{Var}(\bar{m})$ itself: the test then estimates only the small idiosyncratic part of the variance. Overlapping training sets add $\rho\tau^2 > 0$ to $\bar{c}$, which is the mechanism identified by Dietterich [1] and corrected by Nadeau and Bengio [2] for i.i.d. data. Blocking reduces the spatial part of $\bar{c}$, but adjacent blocks of different folds remain correlated, and with $K = 5$ the test has only four degrees of freedom.

Cluster methods.

The block bootstrap and the random-intercept linear mixed model (LMM) treat the G blocks as independent clusters. They capture all within-block covariance, including the fold effects of (2), but ignore covariance between neighbouring blocks, which is non-negligible when the dependence range is comparable with the block size, and they ignore the correlation ρ of fold effects.

Time-only HAC.

Applying the Diebold–Mariano test with a Newey–West variance separately to the series of each site, as is often done for panels of forecasts, handles temporal dependence but ignores the correlation between sites at the same time and the fold effects.

4 Dependence-aware HAC inference

DA-HAC estimates the two components of (3) from the loss differentials of a single blocked cross-validation run. Its steps are summarised in Algorithm 1 and detailed below.

4.1 Fold-centred differentials

Let m_k be the mean of the d_i in fold k and $\tilde{u}_i = d_i - m_{k(i)}$. Under (2), $\tilde{u}_i = u_i - \bar{u}_{k(i)}$ is free of the fold effects, so that the dependence structure of u can be studied without confusion with the training-induced component.

4.2 Variogram-calibrated bandwidths

The key tuning parameter of any HAC estimator is the bandwidth, i.e. the distance beyond which covariances are neglected. Rules of thumb developed for time series [24, 35] do not transfer to space, and the dependence range of the loss differential is in general different from that of the response: it is the range of the *residual* structure that both models fail to capture. We therefore estimate it from $\tilde{u}$ directly. The directional empirical semivariograms

$$\hat{\gamma}_S(h) = \frac{1}{2|N_S(h)|} \sum_{(i,j)\in N_S(h)} (\tilde{u}_i - \tilde{u}_j)^2, \quad N_S(h) = \{(i,j) : t_i = t_j,\ \|s_i - s_j\| \in I_h\},$$

$$\hat{\gamma}_T(\ell) = \frac{1}{2|N_T(\ell)|} \sum_{(i,j)\in N_T(\ell)} (\tilde{u}_i - \tilde{u}_j)^2, \quad N_T(\ell) = \{(i,j) : s_i = s_j,\ |t_i - t_j| \in I_\ell\},$$

are computed over distance and lag bins I_h, I_ℓ up to a maximum lag (by default half the spatial extent and a quarter of the temporal extent). For purely spatial data, $N_S(h)$ uses all pairs. To each we fit the exponential model $\gamma(h) = c_0 + c_1\{1 - \exp(-h/a)\}$ by the weighted least squares of Cressie [36], with weights $|N(h)|/\gamma(h)^2$. The *practical range* $r = 3a$ is the distance at which the correlation of the structured component falls to 5%; it is set to zero when the structured part explains less than 5% of the sill ($c_1 < 0.05(c_0 + c_1)$). The spatial and temporal bandwidths are

$$b_S = \kappa\, \hat{r}_S, \qquad b_T = \kappa\, \hat{r}_T,$$

with $\kappa = 1$ chosen in a pilot study on separate random seeds (Section 5). We also report the *range ratio* $\max_q b_q/L_q$, the largest bandwidth relative to the extent of the domain along the same axis, as a diagnostic: when it approaches one, the sample contains only a handful of effectively independent regions, and no procedure can estimate the variance of $\hat{\Delta}$ reliably from it.

4.3 Kernel estimate of the test-sampling variance

We use the product-Bartlett kernel

$$w_{ij} = \prod_{q=1}^{p} \Big(1 - \frac{|s_{iq} - s_{jq}|}{b_S}\Big)_+ \Big(1 - \frac{|t_i - t_j|}{b_T}\Big)_+, \tag{4}$$

where a zero bandwidth keeps only pairs with identical coordinates on that axis. Each factor is a one-dimensional Bartlett (triangular) kernel, whose Fourier transform is non-negative; the product is therefore a positive semi-definite kernel on $\mathbb{R}^{p+1}$, which guarantees non-negative variance estimates. (A Bartlett kernel applied to the Euclidean distance in two or more dimensions does not have this property.) The test-sampling variance and its within-fold analogues are

$$\hat{V}_u = \frac{1}{n^2} \sum_{i,j} w_{ij} \tilde{u}_i \tilde{u}_j, \qquad \hat{V}_k = \frac{1}{n_k^2} \sum_{i,j \in F_k} w_{ij} \tilde{u}_i \tilde{u}_j. \tag{5}$$

Only pairs with $w_{ij} > 0$ contribute; they are enumerated with a k-d tree in the Chebyshev metric after scaling each coordinate by its bandwidth, at a cost proportional to the number of such pairs.

4.4 Training-induced variance

Under (2), the between-fold variance of fold means exceeds their within-fold test-sampling variance by the variance of the fold effects. We estimate it by the method of moments,

$$\hat{\tau}^2 = \max\Big\{0,\ s_m^2 - \frac{1}{K} \sum_{k=1}^{K} \hat{V}_k \Big\}. \tag{6}$$

Since $\mathbb{E}[s_m^2]$ subtracts the covariance between fold effects (Proposition 1), $\hat{\tau}^2$ targets $\tau^2(1-\rho)$ rather than τ^2. The correlation ρ is not identifiable from one cross-validation run [13]; following Nadeau and Bengio [2] we set it to the test fraction $\rho = n_{\text{test}}/n = 1/K$. Then $V_\eta = \tau^2[1/K + \rho(1 - 1/K)] = \zeta_K \, \tau^2(1-\rho)$ with

$$\zeta_K = \frac{1}{K} + \frac{1}{K-1},$$

which is exactly the Nadeau–Bengio inflation $1/J + n_{\text{test}}/n_{\text{train}}$ with $J = K$ splits. The final variance estimate is

$$\hat{V} = \hat{V}_u + \zeta_K \hat{\tau}^2. \tag{7}$$

Fold-centring removes from $\tilde{u}$ the fold-level average of u itself, so $\hat{V}_u$ is slightly biased downward when the dependence range is comparable to the fold size; because s_m^2 contains that same component while the $\hat{V}_k$ do not, the deficit is transferred to $\hat{\tau}^2$, which partially compensates for it.

Algorithm 1 DA-HAC test for $H_0 : \Delta_n = 0$

Require: loss differentials d_i, folds $k(i)$, coordinates (s_i, t_i), level α, multiplier κ (default 1)

1: $m_k \leftarrow$ mean of d_i over F_k; $\tilde{u}_i \leftarrow d_i - m_{k(i)}$; $\hat{\Delta} \leftarrow n^{-1} \sum_i d_i$
2: Fit exponential models to the spatial and temporal empirical variograms of $\tilde{u}$; $b_S \leftarrow \kappa \hat{r}_S$, $b_T \leftarrow \kappa \hat{r}_T$
3: Enumerate pairs with $w_{ij} > 0$ under the product-Bartlett kernel (4)
4: Compute $\hat{V}_u$ and $\hat{V}_1, \ldots, \hat{V}_K$ by (5)
5: $\hat{\tau}^2 \leftarrow \max\{0, s_m^2 - K^{-1} \sum_k \hat{V}_k\}$; $\hat{V} \leftarrow \hat{V}_u + (1/K + 1/(K-1))\hat{\tau}^2$
6: Compute $\hat{n}_{\text{eff}}$, ν_u and $\hat{\nu}$ by (8), and the range ratio $\max_q b_q / L_q$
7: **return** $p = 2 \Pr(t_{\hat{\nu}} > |\hat{\Delta}|/\sqrt{\hat{V}})$ and the interval $\hat{\Delta} \pm t_{\hat{\nu}, 1-\alpha/2} \sqrt{\hat{V}}$

4.5 Reference distribution

HAC tests are known to over-reject when the effective sample size is small [27], and $\hat{\tau}^2$ rests on only $K - 1$ degrees of freedom. We approximate the distribution of $\hat{V}$ by a scaled χ^2 with Welch–Satterthwaite [37] degrees of freedom

$$\hat{\nu} = \frac{\hat{V}^2}{\hat{V}_u^2/\nu_u + (\zeta_K \hat{\tau}^2)^2/(K-1)}, \qquad \nu_u = \min\Big\{\hat{n}_{\text{eff}} - 1,\ \prod_q \frac{3L_q}{2b_q}\Big\}, \qquad \hat{n}_{\text{eff}} = \frac{n^{-1}\sum_i \tilde{u}_i^2}{\hat{V}_u}, \tag{8}$$

where $\hat{n}_{\text{eff}}$ is truncated to $[2, n]$, the product runs over the spatial and temporal axes with a positive bandwidth, and L_q is the extent of the domain along axis q. The second term of ν_u is the classical approximation to the degrees of freedom of a kernel spectral estimate [35]: for a lag window k of bandwidth b on a record of length L, $\mathrm{Var}(\hat{V})/V^2 \approx 2(b/L)\int k^2$, with $\int k^2 = 2/3$ for the Bartlett window, and the factors multiply for a product kernel. It caps the degrees of freedom when the bandwidth is not small compared with the domain, the regime in which HAC tests are most anti-conservative. The test statistic $T = \hat{\Delta}/\sqrt{\hat{V}}$ is referred to a Student $t_{\hat{\nu}}$ distribution, and the $(1 - \alpha)$ confidence interval is $\hat{\Delta} \pm t_{\hat{\nu}, 1-\alpha/2}\sqrt{\hat{V}}$.

4.6 Properties

Remark 1 (Special cases) If $b_S = b_T = 0$ and $\hat{\tau}^2 = 0$, then $\hat{V} = n^{-2} \sum_i \tilde{u}_i^2$ and DA-HAC reduces to the one-sample t-test on the d_i. If the data have no spatial (or no temporal) coordinate, the corresponding factor of (4) is omitted. Because the kernel is positive semi-definite, $\hat{V} \geq 0$ always.

Remark 2 (Consistency) For fixed bandwidths, $\hat{V}_u$ is a Conley-type spatial HAC estimator [25]. Under increasing-domain asymptotics with mixing conditions and bandwidths that grow with, but more slowly than, the size of the domain, such estimators are consistent [25, 35]. Our bandwidths are data-dependent and our variance includes a component, $\hat{\tau}^2$, estimated from only K folds; we therefore do not claim exact asymptotic validity, and instead evaluate the finite-sample behaviour of the whole procedure by simulation with known truth.

Remark 3 (Scope) DA-HAC requires only the per-observation losses of the two models and their coordinates; it is agnostic to the learners, to the loss and to the blocking scheme. The same construction applies to any additive performance measure (squared error, absolute error, log loss, Brier score, 0–1 loss). Non-additive measures such as the AUC or the F1-score would require a linearisation (influence function) first.

5 Simulation study I: controlled loss-differential fields

The first study isolates the inferential problem from the learning problem. Prediction errors are simulated directly, so that the true expected loss differential is known exactly and the dependence of the d_i is controlled.

5.1 Design

Fields.

Observations lie on a 12×12 grid of sites observed at 24 time points ($n = 3456$). All fields are separable Gaussian fields with unit variance, exponential spatial correlation $\exp(-h/a)$ and AR(1) temporal dependence with coefficient ϕ. Four dependence levels are considered: *none* ($a = 0$, $\phi = 0$), *weak* ($a = 1$, $\phi = 0.3$), *moderate* ($a = 2$, $\phi = 0.6$) and *strong* ($a = 4$, $\phi = 0.8$). The corresponding practical ranges are $3a = 0, 3, 6, 12$ grid units in space and $-3/\log\phi = 0, 2.5, 5.9, 13.4$ time steps; at the strong level the range is as large as the domain.

Losses.

With ε, e_A and e_B three independent fields, the response noise is ε, model A has errors e_A and model B has errors $\sqrt{1+\delta}\, e_B$. Under squared loss,

$$d_i = (\varepsilon_i - e_{A,i})^2 - (\varepsilon_i - \sqrt{1+\delta}\, e_{B,i})^2 + \eta_{k(i)}, \qquad \mathbb{E}[d_i] = -\delta,$$

so that $\Delta_n = -\delta$ exactly. The fold effects $\eta_k = \sigma_\eta(\sqrt{\rho}\, g + \sqrt{1-\rho}\, h_k)$, with $g, h_1, \ldots, h_K$ i.i.d. standard normal, $\sigma_\eta = 0.05$ and $\rho = 0.25$, mimic the training-induced variability of (2), including its correlation across folds. We use $\delta = 0$ (null) and $\delta \in \{0.1, 0.2, 0.4\}$ (alternatives).

Validation and procedures.

The domain is cut into 4×4-site tiles crossed with 6-step time windows ($G = 36$ blocks), which are allocated to $K = 5$ folds with balanced sizes. We compare the paired t-test on fold means, the i.i.d. bootstrap, the block bootstrap and a random-intercept LMM (both with the 36 blocks as clusters), the Diebold–Mariano test with a site-wise Newey–West variance (DM–HAC), and DA-HAC. Three ablations of DA-HAC are included: without the training term ($\hat{\tau}^2 = 0$), with a normal reference distribution, and with a fixed bandwidth equal to the block size instead of the variogram-calibrated one. Bootstraps use 999 resamples. There are 1000 replicates per dependence level under the null and 300 under each alternative, so that the Monte Carlo standard error of a rejection rate of 5% is 0.7 percentage points.

Pilot.

The bandwidth multiplier κ and the reference distribution were chosen in a pilot study with 300 null replicates per level, run on random seeds disjoint from those of the main study. Rejection rates of DA-HAC varied by at most 2.7 percentage points across $\kappa \in \{0.5, 1, 1.5, 2\}$ at any level, with no systematic trend, and we retained $\kappa = 1$. Capping the degrees of freedom by the kernel term of (8) reduced the rejection rate at the strong level from 17.7% to 15.3% without affecting the other levels by more than 1.4 points, and was retained. The pilot results are included in the code repository.

5.2 Results

Type I error.

Table 1 and Fig. 1 report rejection rates under the null. Procedures that ignore dependence fail quickly: the i.i.d. bootstrap rejects a true null in 19% of replicates under weak dependence and in 80% under strong dependence, and the time-only DM–HAC test, which ignores correlation between sites, behaves almost as badly (17% and 70%). The two cluster procedures, block bootstrap and LMM, are much better but still drift upwards as the range of dependence approaches the block size (15% and 14% at the moderate level, 27% and 25% at the strong level), because they treat neighbouring blocks as independent. Even without any spatial or temporal dependence they reject 9% and 7% of true nulls, as they do not account for the correlation of fold effects. The paired t-test on blocked fold means is the best conventional procedure, in line with Proposition 1, since blocking removes most of the covariance between fold means; it nevertheless reaches 11% at the moderate and 18% at the strong level.

DA-HAC has rejection rates of 4.8%, 7.2%, 9.9% and 12.8% at the four levels. It is the procedure closest to the nominal level without dependence and under moderate and strong dependence; under weak dependence the paired t-test (6.7%) is marginally closer. Its ablation with a fixed bandwidth equal to the block size is slightly better at the weak and moderate levels (5.6% and 8.8%). That variant uses knowledge of a sensible block size, which the variogram calibration recovers from the data. DA-HAC is *not* exactly calibrated under moderate and strong dependence. At the strong level the practical range (12 sites and 13 time steps) equals the extent of the domain; the median estimated range ratio b/L is 0.52, the sample contains only a few effectively independent regions, and the estimated ranges are about half the true ones because the variogram cannot be estimated beyond the domain. In this regime no method we studied is reliable, and the range ratio is an informative warning (Section **??**).

Ablations.

Removing the training term $\zeta_K \hat{\tau}^2$ almost triples the rejection rate at the weak and moderate levels (14% and 24% instead of 7% and 10%): fold-centred kernel estimators miss the between-fold component of the variance, whose estimation is therefore essential. Replacing the Satterthwaite reference by a normal distribution adds up to 3 percentage points, the largest increase being at the strong level.

Table 1 Experiment 1: rejection rate (%) of a true null at $\alpha = 0.05$ (1000 replicates per level) and coverage (%) of the 95% confidence interval (all 1900 replicates per level). Bold: rejection rate outside the 95% Monte Carlo band $[3.6, 6.4]$ around 5%.

	Type I error (%)				Coverage (%)			
Procedure	None	Weak	Mod.	Strong	None	Weak	Mod.	Strong
Paired t (folds)	**6.6**	**6.7**	**10.7**	**17.9**	93.5	93.6	89.2	81.4
i.i.d. bootstrap	**8.1**	**19.4**	**57.4**	**80.3**	91.8	79.7	43.0	19.3
Block bootstrap	**9.2**	**8.7**	**15.3**	**27.2**	91.1	90.5	82.6	70.4
Mixed model	**7.4**	**8.0**	**13.6**	**24.5**	92.3	91.7	85.7	74.2
DM–HAC (time only)	**8.4**	**17.0**	**48.2**	**70.2**	91.8	82.4	51.8	28.1
DA-HAC (proposed)	4.8	**7.2**	**9.9**	**12.8**	95.4	92.6	89.7	85.6
DA-HAC without $\hat{\tau}^2$ term	**7.1**	**14.1**	**23.6**	**25.6**	93.1	85.3	75.8	73.2
DA-HAC, normal reference	5.8	**7.8**	**10.8**	**15.6**	94.4	92.1	88.7	82.8
HAC, block-size bandwidth	6.3	5.6	**8.8**	**14.3**	94.1	94.3	90.5	83.6

Coverage, width and power.

Coverage of the 95% intervals over all replicates (null and alternatives, Table 1) mirrors the type I error: 95.4%, 92.6%, 89.7% and 85.6% for DA-HAC, against 79.7%, 43.0% and 19.3% for the i.i.d. bootstrap from the weak level on. The DA-HAC intervals are wider than those of the cluster procedures (Fig. 1b), which is the price of validity: the narrow intervals of the i.i.d. bootstrap and DM–HAC do not widen with dependence at all. Raw rejection rates under the alternatives (Fig. 2) are not comparable across procedures whose sizes differ by an order of magnitude; we therefore also report size-adjusted power, using as critical value the 5% quantile of each procedure's null p-values (Table 2). DA-HAC has about the same size-adjusted power as the paired t-test. Without dependence it is less powerful than the cluster procedures (61% against 81% at $\delta = 0.2$), because the training term is estimated from only $K - 1 = 4$ degrees of freedom; under moderate dependence the cluster procedures retain more size-adjusted power, but this advantage cannot be realised in practice, since their nominal tests over-reject and their null distribution is unknown.

6 Simulation study II: learners on spatio-temporal data

The second study reproduces the full pipeline of an applied comparison: data generation, model fitting by cross-validation and testing. The difficulty in such a study is to know the truth. We obtain an exact null by exchangeability.

Table 2 Experiment 1: size-adjusted power (%) for true loss gaps $\delta \in \{0.1, 0.2, 0.4\}$ (300 replicates each). The critical value of each procedure is the 5% quantile of its own null p-values; it is undefined (–) when more than 5% of null bootstrap p-values equal the smallest attainable value.

	No dependence			Moderate			Strong		
Procedure	0.1	0.2	0.4	0.1	0.2	0.4	0.1	0.2	0.4
Paired t (folds)	24	54	96	7	13	29	5	7	9
i.i.d. bootstrap	31	82	100	–	–	–	–	–	–
Block bootstrap	29	81	100	9	18	46	9	12	17
Mixed model	30	81	100	10	20	49	5	8	12
DM–HAC (time only)	31	83	100	10	16	46	7	8	10
DA-HAC (proposed)	26	61	89	7	14	29	5	7	12

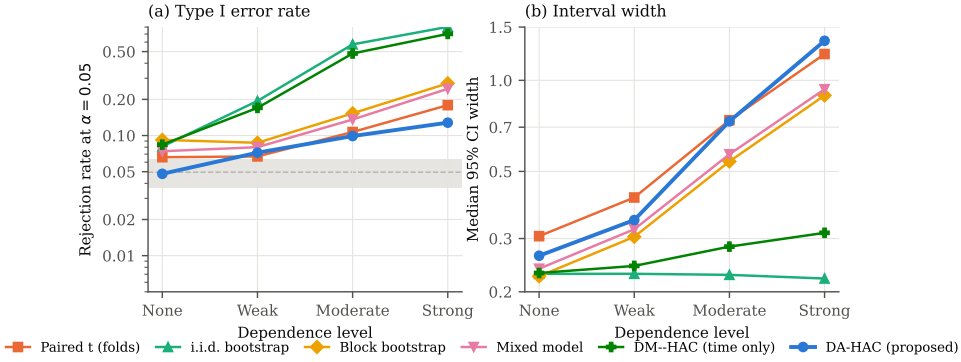


Fig. 1 Experiment 1. (a) Rejection rate of a true null hypothesis (log scale); the dashed line is the nominal level and the grey band the 95% Monte Carlo band. (b) Median width of the 95% confidence interval under the null.

6.1 Design

Data.

On a 10×10 grid observed at 20 time points ($n = 2000$), three covariates x_1, x_2, x_3 and a latent residual w are independent spatio-temporal fields with the dependence levels of Section 5. The response is

$$y = g(x_1) + (1-\gamma)\, g(x_2) + x_3 + w + \epsilon, \qquad g(v) = \sin(2v) + v, \qquad \epsilon \sim N(0, 0.5^2) \text{ i.i.d.}$$

Models.

Model A is XGBoost [38] (100 trees of depth 4, learning rate 0.1, row subsampling 0.8) fitted on (x_1, x_3, s, t); model B is the same learner fitted on (x_2, x_3, s, t). Including the coordinates lets both models exploit the spatial and temporal structure of w, as is common in practice.

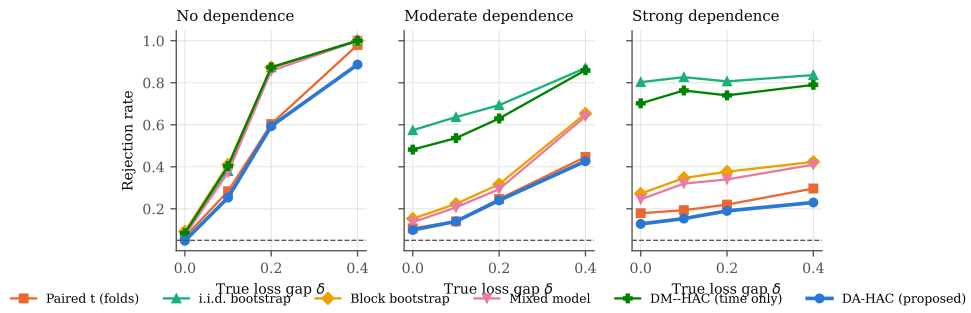


Fig. 2 Experiment 1. Raw rejection rate as a function of the true loss gap δ ($\delta = 0$ is the null). High curves at $\delta = 0$ indicate invalid tests, not power.

Exact null by exchangeability.

When $\gamma = 0$, the joint distribution of the data is invariant under the exchange of x_1 and x_2, which exchanges the roles of A and B. Hence $\mathbb{E}[d_i] = 0$ for every observation and every validation design: the two models have exactly the same expected performance, although in any given sample one of them will appear better. When $\gamma > 0$, x_1 carries more signal than x_2 and A is truly better; we use $\gamma \in \{0.1, 0.2, 0.3\}$.

Validation and procedures.

Two pipelines are applied to each data set. The *conventional* pipeline uses random 5-fold cross-validation with the paired t-test on fold means and the i.i.d. bootstrap, and 5×5 repeated random cross-validation with the corrected resampled t-test [2]. The *dependence-aware* pipeline uses blocked 5-fold cross-validation (5×5-site tiles crossed with 4-step windows, $G = 20$ blocks, and a buffer that removes from the training set observations within one site and one time step of the test fold) with the procedures of Section 5. There are 300 replicates per dependence level under the null and 150 under each alternative.

7 Case studies

7.1 Data and protocol

We use three public data sets obtained from the Rdatasets collection [39].

NYC weather.

Hourly weather records at the three New York City airports (EWR, JFK, LGA) in 2013, distributed with the `nycflights13` package [40] ($n = 26{,}110$ after removing five records with missing temperature, humidity or wind speed). The task is to classify whether precipitation was recorded in the hour, from temperature, dew point, humidity, wind direction and speed, pressure, month, hour and airport; 6.7% of hours are positive. The loss is the logarithmic loss.

Melbourne pedestrians.

Hourly pedestrian counts at four sensors in Melbourne in 2015–2016, distributed with the `tsibble` package [41] (n = 66,037). The task is to predict $\log(1 + \text{count})$ from hour, day of week, month, day of year, year and sensor, under squared loss.

Las Rosas maize.

Yield-monitor data from a maize field in Las Rosas, Argentina, in 1999 and 2001 [42], distributed with the `agridat` package (n = 3443, field of about 660×380 m). The task is to predict yield from the nitrogen rate, brightness value, topographic zone, year and coordinates, under squared loss.

For the two hourly data sets, blocks are two-week windows (26 and 53 blocks); as the airports lie within about 35 km of one another and the sensors within about 2 km, they are treated as co-located, so that all sites observed at the same time may be correlated, and a 24-hour buffer is removed from the training set on each side of a test block. For the maize data, blocks are 75×75 m tiles (35 non-empty blocks) with a 10 m buffer. We compare XGBoost (300 trees of depth 6, learning rate 0.05, row and column subsampling 0.8) with a random forest [43] (200 trees, minimum leaf size 5, 50% of features per split) and with a linear baseline (logistic or ridge regression on standardised, mean-imputed features). All models use scikit-learn [44] or the XGBoost library with default settings otherwise; no hyperparameter was tuned, since our object is the comparison, not the models.

Seed-null diagnostic.

For each data set we also compare XGBoost with itself, the two runs differing only in their random seed (30 pairs). By symmetry, the expected loss differential of such a comparison is zero, while in each run the stochastic subsampling produces a non-zero observed difference. A procedure that frequently declares one seed "significantly better" than the other confuses training noise with a real improvement. Because the data are held fixed, this diagnostic measures the sensitivity of each procedure to training randomness only; it is not a full assessment of size, which Sections 5 and 6 provide.

Acknowledgements. The author thanks the maintainers of the `nycflights13`, `tsibble` and `agridat` R packages and of the Rdatasets collection for making the data used in this study openly available.

Declarations

- **Funding.** This research received no specific grant from any funding agency in the public, commercial or not-for-profit sectors.
- **Conflict of interest.** The author declares no competing interests.
- **Ethics approval and consent to participate.** Not applicable.
- **Consent for publication.** Not applicable.
- **Data availability.** All data sets are public and are redistributed, with their original licences, in the code repository, together with the raw outputs of all experiments.

- **Code availability.** The Python package `dacomp` and all scripts needed to reproduce the simulations, case studies, tables and figures are available at https://github.com/2zalab/Statistics-for-Data-Science-and-AI under the MIT licence.
- **Author contribution.** I.T. conceived the study, developed the methodology and software, carried out the experiments and wrote the manuscript.

References

[1] Dietterich, T.G.: Approximate statistical tests for comparing supervised classification learning algorithms. Neural Computation **10**(7), 1895–1923 (1998)

[2] Nadeau, C., Bengio, Y.: Inference for the generalization error. Machine Learning **52**(3), 239–281 (2003)

[3] Demšar, J.: Statistical comparisons of classifiers over multiple data sets. Journal of Machine Learning Research **7**, 1–30 (2006)

[4] Benavoli, A., Corani, G., Demšar, J., Zaffalon, M.: Time for a change: a tutorial for comparing multiple classifiers through Bayesian analysis. Journal of Machine Learning Research **18**(77), 1–36 (2017)

[5] Roberts, D.R., Bahn, V., Ciuti, S., Boyce, M.S., Elith, J., Guillera-Arroita, G., Hauenstein, S., Lahoz-Monfort, J.J., Schröder, B., Thuiller, W., Warton, D.I., Wintle, B.A., Hartig, F., Dormann, C.F.: Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure. Ecography **40**(8), 913–929 (2017)

[6] Ploton, P., Mortier, F., Réjou-Méchain, M., Barbier, N., Picard, N., Rossi, V., Dormann, C., Cornu, G., Viennois, G., Bayol, N., Lyapustin, A., Gourlet-Fleury, S., Pélissier, R.: Spatial validation reveals poor predictive performance of large-scale ecological mapping models. Nature Communications **11**, 4540 (2020)

[7] Meyer, H., Reudenbach, C., Hengl, T., Katurji, M., Nauss, T.: Improving performance of spatio-temporal machine learning models using forward feature selection and target-oriented validation. Environmental Modelling & Software **101**, 1–9 (2018)

[8] Bergmeir, C., Benítez, J.M.: On the use of cross-validation for time series predictor evaluation. Information Sciences **191**, 192–213 (2012)

[9] Valavi, R., Elith, J., Lahoz-Monfort, J.J., Guillera-Arroita, G.: blockCV: An R package for generating spatially or environmentally separated folds for k-fold cross-validation of species distribution models. Methods in Ecology and Evolution **10**(2), 225–232 (2019)

[10] Linnenbrink, J., Milà, C., Ludwig, M., Meyer, H.: kNNDM CV: k-fold nearest-neighbour distance matching cross-validation for map accuracy estimation.

Geoscientific Model Development **17**, 5897–5912 (2024)

[11] Diebold, F.X., Mariano, R.S.: Comparing predictive accuracy. Journal of Business & Economic Statistics **13**(3), 253–263 (1995)

[12] Hering, A.S., Genton, M.G.: Comparing spatial predictions. Technometrics **53**(4), 414–425 (2011)

[13] Bengio, Y., Grandvalet, Y.: No unbiased estimator of the variance of K-fold cross-validation. Journal of Machine Learning Research **5**, 1089–1105 (2004)

[14] Bouckaert, R.R., Frank, E.: Evaluating the replicability of significance tests for comparing learning algorithms. In: Advances in Knowledge Discovery and Data Mining (PAKDD 2004). Lecture Notes in Computer Science, vol. 3056, pp. 3–12. Springer, ??? (2004)

[15] Bayle, P., Bayle, A., Janson, L., Mackey, L.: Cross-validation confidence intervals for test error. In: Advances in Neural Information Processing Systems, vol. 33, pp. 16339–16350 (2020)

[16] Bates, S., Hastie, T., Tibshirani, R.: Cross-validation: what does it estimate and how well does it do it? Journal of the American Statistical Association **119**(546), 1434–1445 (2024)

[17] Bouthillier, X., Delaunay, P., Bronzi, M., Trofimov, A., Nichyporuk, B., Szeto, J., Mohammadi Sepahvand, N., Raff, E., Madan, K., Voleti, V., Ebrahimi Kahou, S., Michalski, V., Arbel, T., Pal, C., Varoquaux, G., Vincent, P.: Accounting for variance in machine learning benchmarks. Proceedings of Machine Learning and Systems **3**, 747–769 (2021)

[18] Varoquaux, G.: Cross-validation failure: small sample sizes lead to large error bars. NeuroImage **180**, 68–77 (2018)

[19] Le Rest, K., Pinaud, D., Monestiez, P., Chadoeuf, J., Bretagnolle, V.: Spatial leave-one-out cross-validation for variable selection in the presence of spatial autocorrelation. Global Ecology and Biogeography **23**(7), 811–820 (2014)

[20] Schratz, P., Muenchow, J., Iturritxa, E., Richter, J., Brenning, A.: Hyperparameter tuning and performance assessment of statistical and machine-learning algorithms using spatial data. Ecological Modelling **406**, 109–120 (2019)

[21] Wadoux, A.M.J.-C., Heuvelink, G.B.M., Bruin, S., Brus, D.J.: Spatial cross-validation is not the right way to evaluate map accuracy. Ecological Modelling **457**, 109692 (2021)

[22] Bergmeir, C., Hyndman, R.J., Koo, B.: A note on the validity of cross-validation for evaluating autoregressive time series prediction. Computational Statistics &

Data Analysis **120**, 70–83 (2018)

[23] Harvey, D., Leybourne, S., Newbold, P.: Testing the equality of prediction mean squared errors. International Journal of Forecasting **13**(2), 281–291 (1997)

[24] Newey, W.K., West, K.D.: A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. Econometrica **55**(3), 703–708 (1987)

[25] Conley, T.G.: GMM estimation with cross sectional dependence. Journal of Econometrics **92**(1), 1–45 (1999)

[26] Driscoll, J.C., Kraay, A.C.: Consistent covariance matrix estimation with spatially dependent panel data. Review of Economics and Statistics **80**(4), 549–560 (1998)

[27] Kiefer, N.M., Vogelsang, T.J.: A new asymptotic theory for heteroskedasticity-autocorrelation robust tests. Econometric Theory **21**(6), 1130–1164 (2005)

[28] Künsch, H.R.: The jackknife and the bootstrap for general stationary observations. The Annals of Statistics **17**(3), 1217–1241 (1989)

[29] Politis, D.N., Romano, J.P.: The stationary bootstrap. Journal of the American Statistical Association **89**(428), 1303–1313 (1994)

[30] Lahiri, S.N.: Resampling Methods for Dependent Data. Springer, New York (2003)

[31] Giudici, P., Raffinetti, E., Riani, M.: Robust machine learning models: linear and nonlinear. International Journal of Data Science and Analytics, 1043–1050 (2024) https://doi.org/10.1007/s41060-024-00512-1

[32] Chicco, D., Oneto, L.: Eleven quick tips to reduce overfitting in machine learning. International Journal of Data Science and Analytics (2026) https://doi.org/10.1007/s41060-026-01246-y

[33] Caponero, J., Costa, L.C., Ara, A.: Beyond significance: why the d-value matters more in big data contexts. International Journal of Data Science and Analytics (2025) https://doi.org/10.1007/s41060-025-00903-y

[34] Gneiting, T., Raftery, A.E.: Strictly proper scoring rules, prediction, and estimation. Journal of the American Statistical Association **102**(477), 359–378 (2007)

[35] Andrews, D.W.K.: Heteroskedasticity and autocorrelation consistent covariance matrix estimation. Econometrica **59**(3), 817–858 (1991)

[36] Cressie, N.: Fitting variogram models by weighted least squares. Journal of the International Association for Mathematical Geology **17**(5), 563–586 (1985)

[37] Satterthwaite, F.E.: An approximate distribution of estimates of variance components. Biometrics Bulletin **2**(6), 110–114 (1946)

[38] Chen, T., Guestrin, C.: XGBoost: A scalable tree boosting system. In: Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, pp. 785–794 (2016)

[39] Arel-Bundock, V.: Rdatasets: A collection of datasets originally distributed in various R packages. https://vincentarelbundock.github.io/Rdatasets/ (2024)

[40] Wickham, H.: nycflights13: Flights that Departed NYC in 2013. R package, https://CRAN.R-project.org/package=nycflights13 (2021)

[41] Wang, E., Cook, D., Hyndman, R.J.: A new tidy data structure to support exploration and modeling of temporal data. Journal of Computational and Graphical Statistics **29**(3), 466–478 (2020)

[42] Anselin, L., Bongiovanni, R., Lowenberg-DeBoer, J.: A spatial econometric approach to the economics of site-specific nitrogen management in corn production. American Journal of Agricultural Economics **86**(3), 675–687 (2004)

[43] Breiman, L.: Random forests. Machine Learning **45**(1), 5–32 (2001)

[44] Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., Vanderplas, J., Passos, A., Cournapeau, D., Brucher, M., Perrot, M., Duchesnay, E.: Scikit-learn: Machine learning in Python. Journal of Machine Learning Research **12**, 2825–2830 (2011)